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NPTEL (https://swayam.gov.in/explorer?ncCode=NPTEL) » The Joy of Computing using Python (course)

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Course outline

About NPTEL ()

How does an NPTEL online course work? ()

Week 0 ()

Week 1 ()

Week 2 ()

Week 3 ()

Week 3 : Assignment 3

Assignment not submitted **Due date: 2024-02-14, 23:59 IST.**

1) Consider the following code snippet: ***1 point***

```
elements = []
n = int(input("Enter the number of elements: "))
for i in range(n):
    item = input("Enter element " + str(i + 1) + ": ")
    elements.append(item)
print("The entered elements are:", elements)
```

What does the code do?

- ○ Takes a list of numbers as input, appends each number to a list, and prints the list of numbers
- ○ Takes a list of elements as input, appends each element to a list, and prints the list of elements
- ◉ Takes the count of elements followed by the same number of elements as input, appends each element to a list, and prints the list of elements
- ○ Takes the count of elements, appends each element to a list, and prints the count of elements entered.

2) Data Structure is a way by which you can organize/arrange your data. Which of the following statements are true about List Data Structure: [MSQ] ***1 point***

- ☑ It is a flexible Data Structure
- ☑ Elements could be added to a list.

Lists Part 1 : Introduction (unit?unit=57&lesson=58)
Lists Part 2 : Manipulation (unit?unit=57&lesson=59)
Lists Part 3 : Operations (unit?unit=57&lesson=60)
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Loops and Conditionals : Fizzbuzz 01 (unit?unit=57&lesson=62)
Loops and Conditionals : Fizzbuzz 02 (unit?unit=57&lesson=63)
Crowd Computing - Just estimate 01 (unit?unit=57&lesson=64)
Crowd Computing - Just estimate 02 (unit?unit=57&lesson=65)
Crowd Computing - Just estimate 03 (unit?

- [x] Elements could be subtracted from a list.
- [x] **This_is_not_List = [ ]** is an empty list

3) Consider the following code snippet: ***1 point***

```
L = ['Amar', 'Akbar', 'Anthony']
print(L[2], 'is brother of', L[1], 'and', L[0])
```

What does the above code print?

- ○ Amar, Akbar, Anthony
- ○ amar is brother of anthony and akbar
- ○ Amar is brother of Anthony and Akbar
- ◉ Anthony is brother of Akbar and Amar

4) Consider the following code snippet: ***1 point***

```
numbers = [2, 4, 6, 8, 10]
for i in range(len(numbers)):
    numbers[i] *= 2
print("Updated list:", numbers)
```

What does the code do?

- ◉ Takes a list of numbers, multiplies each number by 2, and prints the updated list
- ○ Takes a list of numbers, appends each number to the list twice, and prints the updated list
- ○ Takes a list of numbers, removes even numbers from the list, and prints the updated list
- ○ Takes a list of numbers, divides each number by 2, and prints the updated list

5) What will be the output of the following Python code? ***1 point***

```
import random
def genetic_evolution(binary_string):
    mutation_probability = 0.1  # Adjust this probability as needed
    evolved_string = ""
    for bit in binary_string:
        if random.random() < mutation_probability:
            evolved_string += "1" if bit == "0" else "0"
        else:
            evolved_string += bit
    return evolved_string
initial_string = "0000000000"
result = genetic_evolution(initial_string)
print(result)
```

- ○ "1111111111"

unit=57&lesson=66)
Crowd Computing - Just estimate 04 (unit?unit=57&lesson=67)
Crowd Computing - Just estimate 05 (unit?unit=57&lesson=68)
Crowd Computing - Just estimate 06 (unit?unit=57&lesson=69)
Permutations - Jumbled Words 01 (unit?unit=57&lesson=70)
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Permutations - Jumbled Words 03 (unit?unit=57&lesson=72)
Theory of Evolution 01 (unit?unit=57&lesson=73)
Theory of Evolution 02 (unit?unit=57&lesson=74)
Theory of Evolution 03

- "0000000000"
- (selected) A string with some 1s and some 0s
- The function will raise an error

6) Consider the following code snippet: ***1 point***

```
numbers = []
n = int(input("Enter the number of elements: "))
for i in range(n):
    num = int(input("Enter number " + str(i + 1) + ": "))
    numbers.append(num)
result = sum(numbers)
average = result / n
print("Sum of the numbers:", result)
print("Average of the numbers:", average)
```

What does the code do?

- (selected) Takes a list of numbers as input, computes the sum of the numbers, and prints the sum along with the average
- Takes the count of elements, computes the sum of the elements, and prints the sum along with the average
- Takes a list of numbers as input, computes the average of the numbers, and prints the average along with the sum
- None of the above

7) What will be the output of the following Python code? ***1 point***

```
import random
def permutation_example(word):
    word_list = list(word)
    random.shuffle(word_list)
    shuffled_word = ''.join(word_list)
    return shuffled_word
input_word = "python"
result = permutation_example(input_word)
print(result)
```

- "Python"
- (selected) A random permutation of the letters in "python"
- "random"
- The function will raise an error

8) Consider the following code snippet: ***1 point***

(unit?unit=57&lesson=75)

Theory of Evolution 04 (unit?unit=57&lesson=76)

Week 3 Feedback Form: The Joy of Computing using Python (unit?unit=57&lesson=77)

Quiz: Week 3 : Assignment 3 (assessment?name=396)

1. Programming Assignment | Week 3 (/noc24_cs57/progassignment?name=397)

2. Programming Assignment | Week 3 (/noc24_cs57/progassignment?name=398)

3. Programming Assignment | Week 3 (/noc24_cs57/progassignment?name=399)

week 4 ()

Text Transcripts ()

```
numbers = [1, 2, 3, 4, 5]
new_numbers = []
for num in numbers:
    new_numbers.insert(0, num) #insert() is a function that inserts a value at a given index
print("Updated list:", new_numbers)
```

What does the code do? [MSQ]

- [ ] Reverses the numbers list and stores it in new_numbers.
- [x] Creates an empty list named new_numbers and appends elements from numbers to it in reverse order.
- [ ] Produces an error due to an invalid function used for list manipulation.
- [ ] Generates a new list new_numbers with elements from numbers in the same order as numbers.

9) Which of the following are the examples of Social Computing/Crowd Computing: [MSQ] ***1 point***

- [x] Wikipedia
- [x] Stack Exchange
- [x] Quora
- [x] Facebook

10) Consider the following code snippet: ***1 point***

```
for i in range(1, 21):
    if i % 3 == 0 and i % 5 == 0:
        print("FizzBuzz")
    elif i % 3 == 0:
        print("Fizz")
    elif i % 5 == 0:
        print("Buzz")
    else:
        print(i)
```

What does the code do?

- ( ) Prints numbers from 1 to 20, replacing multiples of 3 with "Fizz", multiples of 5 with "Buzz", and multiples of both 3 and 5 with "FizzBuzz".
- (•) Prints numbers from 1 to 20, replacing multiples of 3 and 5 with "FizzBuzz", multiples of 3 with "Fizz", and multiples of 5 with "Buzz".
- ( ) Prints numbers from 1 to 20, replacing multiples of 3 with "Fizz", multiples of 5 with "Buzz", and numbers divisible by both 3 and 5 with their product.
- ( ) Prints numbers from 1 to 20, replacing multiples of 3 with "Buzz", multiples of 5 with "Fizz", and numbers divisible by both 3 and 5 with "FizzBuzz".

You may submit any number of times before the due date. The final submission will be consid for grading.

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